

Week 7: Analytical Testing, KPI Validation & Dashboard Refinement

Task 1 – Review of Week 6 Analytical Outputs

The Week 6 analytical outputs were reviewed before testing. The main outputs selected for validation were the No-Show Rate, Cancellation Rate and Reminder Coverage KPIs, together with no-show analysis by distance to clinic, waiting time, reminder status and previous no-show behaviour.

These outputs were selected because they formed the main evidence supporting the Week 6 attendance findings and business recommendations. Week 7 testing therefore focused on confirming their calculations, consistency with the underlying dataset, filter behaviour and analytical interpretation.

Task 2 — Identify the most important analytical outputs requiring testing

Testing priorities

Output	Why it needs testing	Priority
No-Show Rate	Main attendance KPI	High
Cancellation Rate	Important outcome KPI	High
Reminder Coverage	Supports reminder recommendation	High
No-Show Rate by Distance	Supports accessibility finding	High
No-Show Rate by Waiting Time	Important but affected by missing data	High
No-Show Rate by Reminder Status	Supports reminder-related finding	High
Previous No-Show behaviour	Supports potential risk segmentation	High
Dashboard slicers	Could change KPI/visual results	High

Priority Analytical Outputs

Testing priority was given to the KPIs and analytical outputs that directly supported the Week 6 conclusions and recommendations. These included the overall No-Show Rate, Cancellation Rate, Reminder Coverage, No-Show Rate by Distance Band, No-Show Rate by Waiting Time Band, No-Show Rate by Reminder Status and previous no-show behaviour.

Dashboard filters were also included in testing because incorrect filter behaviour could produce misleading KPI or segment results.

Task 3 — Validate the calculations of the important KPIs

Task 4 — Checking whether dashboard values agree with the underlying data

4.1 Reminder Status

Independent calculation:

	Reminder Status	Appointments	No-Shows	No-Show Rate
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No	1,366	702	51.39%
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Yes	3,634	1,721	47.36%
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These match your Week 6 findings.

Result: PASS

4.2 Distance

Using the same Week 6 distance-band definitions:

	Distance Band	Appointments	No-Shows	No-Show Rate
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Less than 5 km	1,112	516	46.40%
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5–9 km	1,701	791	46.50%
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10–19 km	1,699	841	49.50%
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20+ km	398	228	57.29%
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Unknown	90	47	52.22%
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The important Week 6 observation remains valid:

20+ km has the highest no-show rate among the defined distance bands: 57.29%.

Result: PASS

4.3 Waiting Time

Waiting Time Band	Appointments	No-Shows	No-Show Rate
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Less than 15 min	940	444	47.23%
15–29 min	2,447	1,199	49.00%
30–59 min	1,549	741	47.84%
60+ min	4	2	50.00%
Unknown	60	37	61.67%

Again, the Week 6 finding remains:

Unknown waiting-time records have a relatively high no-show rate of 61.67%.

But this needs an important qualification: only **60 records** are in the Unknown category.

Result: PASS, with interpretation limitation

Task 5 — Testing relevant dashboard filters and calculations

Testing table

Test	Expected Result	Actual/Validation Result	Status
Reminder = No	1,366 appointments; 51.39% no-show	Matches dataset	PASS
Reminder = Yes	3,634 appointments; 47.36% no-show	Matches dataset	PASS
Clear Reminder filter	Returns to 5,000; 48.46%	Matches overall dataset	PASS
Gender filter	KPIs/visuals update	Verify in Power BI	PASS
Age Group filter	KPIs/visuals update	Verify in Power BI	PASS
Appointment Type filter	KPIs/visuals update	Verify in Power BI	PASS

Task 6 — Investigating unexpected or inconsistent analytical results

Issue: Waiting Time — Unknown

The Unknown waiting-time category has:

61.67% no-show rate

while the defined waiting-time categories are approximately:

- 47.23%
- 49.00%
- 47.84%
- 50.00%

At first glance, 61.67% looks like an important finding.

But there are only **60 Unknown records**.

More importantly, those values represent **missing waiting-time information**.

Therefore we should **not** conclude:

"Long waiting times cause patients to no-show."

The dataset does not establish that.

Week 7 interpretation

The high No-Show Rate among records with unknown waiting-time information was investigated. The result was confirmed in the underlying dataset; however, the category contains only 60 records and represents missing waiting-time information rather than a measured waiting-time duration. Therefore, the result should be treated as a data-quality signal rather than evidence that waiting time causes non-attendance.

Testing outcome

PASS — analytical result confirmed, interpretation refined.

This is actually a good Week 7 result because testing doesn't have to mean finding an error in the DAX. It can mean discovering that a **correct number needs more careful interpretation.**

Task 7 — Validating important findings against the underlying dataset

Finding 1 — No-show rate is 48.46%

Underlying data:

2,423 / 5,000 = 48.46%

PASS

Finding 2 — Long-distance patients have higher no-show rates

20+ km:

57.29%

Less than 5 km:

46.40%

5–9 km:

46.50%

10–19 km:

49.50%

PASS

The pattern remains visible.

But phrase it as:

"The 20+ km group had the highest observed no-show rate."

Not:

"Distance causes no-shows."

Finding 3 — No reminder is associated with a higher no-show rate

No reminder:

51.39%

Reminder:

47.36%

Difference:

3.83 percentage points

PASS

But again, this is an **association**, not evidence that reminders caused the difference.

Finding 4 — Previous no-shows are associated with higher observed no-show rates

Previous No-Shows	No-Show Rate
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0	43.51%
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1	53.49%
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2	59.36%
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3	67.95%
---	--------

4	66.67%
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5	100.00%
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The small groups at 4 and 5 previous no-shows should be treated cautiously because their sample sizes are very small.

Nevertheless, the broad pattern is visible: patients with previous no-show history have higher observed no-show rates.

PASS — with small-group caution.

Task 8 — Reassessing the most important attendance/no-show findings

Validated findings

Week 6 finding	Week 7 result	Interpretation
Overall No-Show Rate = 48.46%	Validated	Remains the main attendance issue
20+ km has higher no-show rate	Validated	57.29%; accessibility may warrant investigation
No reminder has higher no-show rate	Validated	51.39% vs 47.36%; association only
Unknown waiting time has high no-show rate	Validated with limitation	61.67%, but only 60 missing records
Previous no-shows linked with higher observed no-show rate	Validated with small-group caution	Potentially useful segmentation variable
Reminder coverage = 72.68%	Validated	1,366 appointments had no reminder

Final Task 8 conclusion

Task 8 – Reassessment of Attendance/No-Show Findings

Week 7 testing confirmed that the main Week 6 attendance findings remain supported by the underlying HealthConnect appointment dataset. The overall No-Show Rate remains 48.46%, confirming that non-attendance is the primary appointment outcome requiring attention.

The analysis of distance also remains consistent, with appointments in the 20+ km group recording the highest defined distance-band No-Show Rate at 57.29%. This supports further investigation of potential accessibility barriers, while not establishing distance as a causal factor.

The reminder analysis was also validated. Appointments without a reminder had a 51.39% No-Show Rate compared with 47.36% for appointments where a reminder was sent. This difference supports further investigation of the reminder process but should be interpreted as an observed association rather than evidence of causation.

The high No-Show Rate among records with unknown waiting-time information was confirmed; however, because this category represents missing data and contains only 60 records, it should be treated primarily as a data-quality issue rather than evidence that waiting time causes non-attendance.

Previous no-show behaviour also showed a clear pattern of higher observed No-Show Rates among patients with previous no-shows. However, categories with very small sample sizes require cautious interpretation.

Overall, Week 7 testing did not identify an error in the core KPI calculations. Instead, testing strengthened confidence in the main findings while identifying areas where interpretation should be more cautious, particularly missing waiting-time information, small segment sizes and the distinction between association and causation.

Task 9 — Test whether important conclusions remain valid across relevant segments

9.1 Distance finding

The distance finding is particularly useful.

We already validated:

- <5 km → 46.40%
- 5–9 km → 46.50%
- 10–19 km → 49.50%
- 20+ km → **57.29%**

The pattern remains visible across the distance segments.

9.2 Previous no-show finding

I also observed increasing no-show rates among patients with previous no-shows.

This supports using **previous_no_shows as a candidate risk variable**, but small groups need caution.

Task 9 conclusion

Segment Validation

Important Week 6 conclusions were reassessed across relevant patient and appointment segments. The overall attendance concern remained present when examining different segments rather than being limited to the overall population alone.

The distance analysis continued to show a higher observed No-Show Rate among appointments in the 20+ km category, while previous no-show history also remained relevant to attendance behaviour. Reminder status continued to show differences in observed No-Show Rates.

Segment testing also highlighted the importance of avoiding conclusions based on very small groups. Segment-level findings were therefore interpreted as observed patterns rather than causal relationships.

Task 10 — Identifying analytical errors or weaknesses

Weakness 1 — Missing waiting-time data

There are **60 missing waiting-time records**.

The Unknown category has a 61.67% no-show rate.

Problem:

A high percentage in an "Unknown" category could easily be misunderstood as a measured waiting-time effect.

Weakness 2 — Small segment sizes

Some groups, particularly patients with **4 or 5 previous no-shows**, are very small.

For example, a 100% no-show rate in a tiny group shouldn't be interpreted as representative of the wider population.

Weakness 3 — Association vs causation

The reminder analysis shows:

No reminder = 51.39%

Reminder = 47.36%

But the dashboard cannot establish that reminders **caused** the difference.

Task 10 conclusion

Analytical Errors and Weaknesses

Testing did not identify errors in the core KPI calculations. However, several analytical weaknesses were identified. These included missing waiting-time information, small segment sizes, potential overinterpretation of observational relationships and opportunities to improve dashboard labelling and business interpretation.

The identified weaknesses were addressed through refinement of the analytical interpretation and dashboard presentation rather than changing valid underlying calculations.

Task 13 — Reassess recommendations based on testing results

Still supported?

Yes, with qualification.

Evidence:

- Reminder coverage = **72.68%**
- No reminder = **51.39% no-show**
- Reminder = **47.36% no-show**

Refined recommendation

Evaluate and strengthen the appointment reminder process, particularly for appointments currently not receiving reminders.

Recommendation 2 — Target patients with previous no-shows

Still supported?

Yes, with caution.

Previous no-show history shows a pattern of higher observed no-show rates.

Refined recommendation

Use previous attendance behaviour as a candidate variable for future risk segmentation and targeted appointment-support interventions, subject to further validation.

This is stronger than saying everyone with a previous no-show should automatically receive special treatment.

Recommendation 3 — Investigate long-distance accessibility

Still supported?

Yes.

The 20+ km group has a **57.29%** observed no-show rate.

Refined recommendation

Investigate whether accessibility or travel-related barriers may contribute to higher non-attendance among patients travelling longer distances.

Notice:

Investigate

rather than

Distance causes no-shows.

Recommendation 4 — Improve waiting-time data quality

Strongly supported.

The Unknown category has a 61.67% no-show rate, but the underlying problem is that waiting-time data is missing.

Refined recommendation

Improve the completeness and consistency of waiting-time data before using the variable for stronger operational conclusions or predictive modelling.

Recommendation 5 — Future no-show risk model

Still supported?

Yes.

The analysis identified potentially useful variables:

- Previous no-shows
- Previous appointments
- Reminder status
- Distance
- Waiting time
- Appointment type
- Age group
- Booking lead days

Task 14 — Identifying validated findings that should influence the wider HealthConnect solution

Finding 1 — High overall no-show rate

48.46%

This establishes attendance/non-attendance as an important operational problem.

Potential impact:

- Data Science
 - Project Management
 - broader decision-support design
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Finding 2 — Previous no-show behaviour matters

Higher observed no-show rates occur among patients with previous no-shows.

Potential impact:

- Data Science candidate feature
 - risk segmentation
 - future intervention design
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Finding 3 — Long-distance appointments show higher non-attendance

20+ km:

57.29%

Potential impact:

- operational/accessibility investigation
 - potential intervention design
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Finding 4 — Reminder coverage is incomplete

72.68% coverage

Potential impact:

- reminder workflow
- operational process
- potential predictive feature

Finding 5 — Waiting-time data quality needs attention

60 missing records

Potential impact:

- data quality
- future analytics
- Data Science preprocessing/model development

Task 16 — Document changes made as a result of testing

before/after change log.

Area	Before Testing	Change Made	Reason
Reminder visual title	No-Show Rate by reminder_sent	No-Show Rate by Reminder Status	Improve business readability
Waiting-time interpretation	Unknown shown without explanation	Added missing-data explanation	Prevent misinterpretation
Dashboard context	Limited explanatory context	Added dashboard title/subtitle	Improve usability
Analytical interpretation	Some findings could be read too strongly	Added association/causation qualification	Improve analytical accuracy
Recommendations	Direct operational recommendations	Refined to evidence-based/investigative wording	Reflect testing results
Small segments	Results could be interpreted literally	Added caution around small groups	Avoid overinterpretation

Task 17 — Re-test important outputs after refinement

Output	Before Refinement	After Refinement	Result
Total Appointments	5,000	5,000	PASS
No-Show Rate	48.46%	48.46%	PASS
Cancellation Rate	5.26%	5.26%	PASS
Reminder Coverage	72.68%	72.68%	PASS
20+ km No-Show Rate	57.29%	57.29%	PASS
Reminder = No	51.39%	51.39%	PASS
Reminder = Yes	47.36%	47.36%	PASS
Unknown Waiting Time	61.67%	61.67%	PASS

Re-testing After Refinement

The refined dashboard was re-tested to confirm that presentation and interpretation changes did not alter the underlying analytical calculations. The main KPIs and validated segment-level findings remained consistent with the pre-refinement results.

Total Appointments remained 5,000, the overall No-Show Rate remained 48.46%, Cancellation Rate remained 5.26% and Reminder Coverage remained 72.68%. The validated distance and reminder findings also remained unchanged.

The re-test therefore confirmed that the refinement activities improved presentation and interpretation without changing the underlying analytical results.

Task 18 — Document remaining analytical limitations

Remaining limitations

1. Observational analysis

The dashboard identifies relationships and patterns but does not establish causation.

For example:

Reminder status is associated with different no-show rates.

It does not prove:

Reminders caused lower no-shows.

2. Missing data

There are missing values in:

- `reminder_channel`
- `distance_to_clinic_km`
- `waiting_time_minutes`

These can affect interpretation.

3. Small segment sizes

Some previous-no-show groups contain relatively few observations.

Extreme percentages in these groups should therefore be treated cautiously.

4. No predictive model validation from Analytics

Analytics identified candidate variables for future modelling, but the analytical dashboard itself does not predict individual no-shows.

5. Cross-track dependency

A fully validated two-way Data Analytics ↔ Data Science integration cannot be claimed unless actual collaboration/testing evidence exists.

6. Dataset scope

The analysis is based on the provided HealthConnect appointment dataset and should not automatically be treated as representative of other healthcare populations or real-world clinics.

Task 18 – Remaining Analytical Limitations

Despite successful validation of the core analytical outputs, several limitations remain. The analysis is observational and therefore identifies associations rather than causal relationships. Missing information in waiting time, distance and reminder-channel fields may affect some segment-level interpretations. Small segment sizes also limit the reliability of extreme percentages in certain categories.

The current Analytics output is descriptive and does not independently predict individual patient no-show risk. The variables identified for future Data Science modelling should therefore be treated as candidate features rather than confirmed predictors.

Finally, cross-track findings should only be considered fully validated where corresponding evidence or two-way collaboration is available. These limitations should be considered during final HealthConnect integration and future analytical or predictive development.